



Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Collected SpaceX launch data via API and web scraping.
 - Performed data wrangling and cleaning using Python (Pandas, NumPy).
 - Conducted exploratory data analysis using visualizations and SQL queries.
 - Built interactive visual analytics with Folium (maps) and Plotly Dash (dashboard).
 - Developed and evaluated several classification models to predict first-stage reuse.
- Summary of all results
 - Identified key factors influencing landing success: payload mass, orbit type, and launch site.
 - Best classification model achieved 88.75% accuracy, enabling confident prediction of first-stage reuse.
 - Dashboards and maps support operational insights for strategic planning.

Introduction

- Project background and context
 - SpaceX's innovation in reusable rockets has changed the space launch market.
 - A new company, Space Y, seeks to compete by understanding and predicting key operational and cost factors.
- Problems you want to find answers
 - Can we estimate the cost of a SpaceX launch?
 - Can we predict whether the first-stage will be reused?
 - How do launch site, payload, orbit, and other variables influence outcomes?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:

- SpaceX REST API (api.spacexdata.com/v4)

- Used to collect structured data including payload, orbit, rocket ID, launch outcome, and reuse status.

- Wikipedia Web Scraping

Extracted historical data from:

https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches

Parsed tabular data on launch dates, mission outcomes, booster versions, and payload masses

Methodology

Executive Summary

- Perform data wrangling
 - Merged and cleaned API + scraped datasets
 - Handled missing data, normalized payloads, encoded categorical variables
- Perform exploratory data analysis (EDA) using visualization and SQL
 - Used visual tools (Matplotlib, Seaborn)
 - SQL queries to uncover payload trends, launch outcomes, and orbit distributions
- Perform interactive visual analytics using Folium and Plotly Dash
 - Folium Maps: To explore launch site geolocation and outcome
 - Plotly Dash: To build a responsive dashboard showing success rates and payload outcomes

Methodology

Executive Summary

- Perform predictive analysis using classification models
 - Trained multiple classification models (Logistic Regression, KNN, etc.)
 - Tuned hyperparameters, evaluated performance using accuracy, precision
 - Best model selected to predict first-stage reuse

Data Collection

- Used SpaceX REST API to gather structured launch data.
- Used web scraping (BeautifulSoup) to extract historical launch data from Wikipedia.
- Stored and managed datasets locally for integration and cleaning.

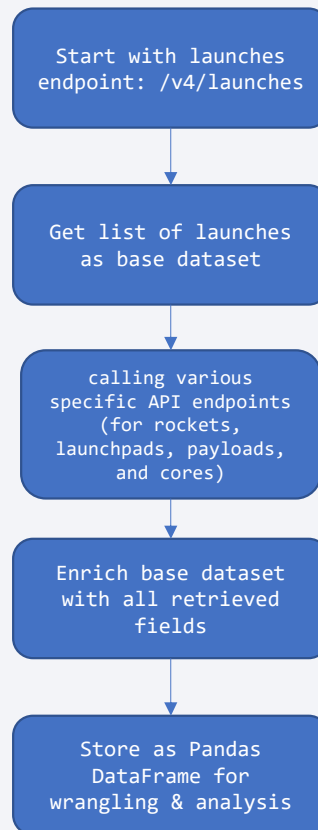
Data Collection – SpaceX API

- Base Data:

- Retrieved from /v4/launches endpoint → includes rocket ID, launchpad ID, payload IDs, core info.
- Rocket Info (/v4/rockets/<rocket_id>): → Booster version name
- Launch Site Info (/v4/launchpads/<launchpad_id>): → Site name, longitude, latitude
- Payload Info (/v4/payloads/<payload_id>): → Payload mass (kg), orbit type
- Core Info (/v4/cores/<core_id>): → Landing outcome, reuse count, serial, flight number
- Final dataset assembled with all details from API into a single structured DataFrame for further wrangling and analysis.

- Source Code:

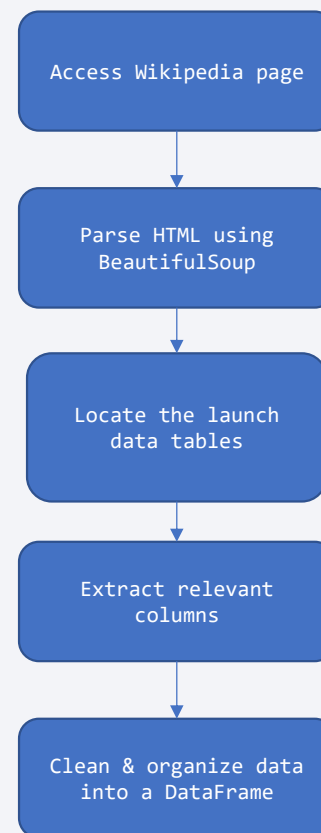
https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- **Target Wikipedia Page:** Fetched HTML content from the specified Wikipedia URL snapshot using requests.
- **Parse HTML:** Utilized BeautifulSoup to parse the HTML, making it navigable for data extraction.
- **Identify Launch Table:** Located the main launch records table within the HTML structure.
- **Extract Data:** Systematically extracted column headers (<th>) and row data (<td>) for each launch, including flight number, date, time, booster version, launch site, payload, payload mass, orbit, customer, launch outcome, and booster landing. Custom helper functions were used for data cleaning (e.g., date_time, get_mass).
- **Build DataFrame:** Transformed the extracted data into a structured Pandas DataFrame.
- **Source Code:**

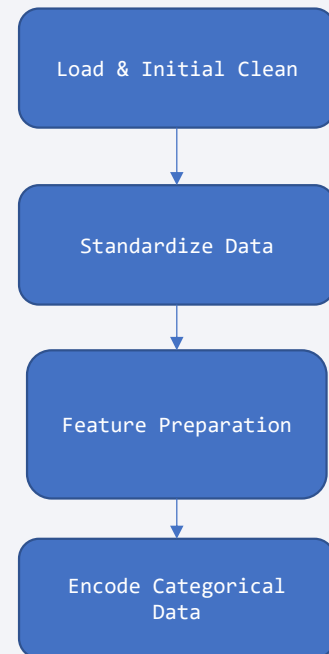
https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856cc/Applied_Data_Science_Capstone/jupyter-labs-webscraping.ipynb



Data Wrangling

- **Load & Initial Clean:** Loaded raw data and performed initial cleaning, including addressing missing values (e.g., 'Payload Mass' zeros) and correcting inconsistencies in text fields.
- **Standardize Data:** Standardized various categorical entries such as Landing Outcome, Booster Version, Orbit, and Customer for uniformity.
- **Feature Preparation:** Converted data types (e.g., 'Date' to datetime) and engineered features as needed.
- **Encode Categorical Data:** Applied one-hot encoding to all categorical variables, converting them into a numerical format suitable for our classification models.
- **Source Code:**

https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/labs-jupyter-spacex-Data%20wrangling.ipynb



EDA with Data Visualization

We performed extensive Exploratory Data Analysis (EDA) using Seaborn and Matplotlib to uncover patterns and relationships within the SpaceX launch data, crucial for predicting first-stage landing success.

- Key Visualizations & Insights:

- Flight Number vs. Payload Mass (Scatter Plot): Revealed that as flight numbers increase, successful landings become more frequent, even with heavier payloads.
- Launch Site Analysis (Scatter & Bar Plots): Showed that CCAFS SLC 40 and KSC LC 39A consistently have higher success rates across various payload masses compared to VAFB SLC 4E.
- Orbit Analysis (Scatter & Bar Plots): Highlighted varying success rates across different orbits. LEO, ISS, ES-L1, SSO, and VLEO orbits generally exhibit high success, while GTO success improved over time.
- Success Rate Over Time (Line Plot): Demonstrated a clear trend of increasing success rates as the number of flights progressed.
- Feature Correlation (Heatmap): Identified key features (FlightNumber, PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount) that are highly correlated with landing success (Class), guiding our feature selection for modeling.

- Source Code:

https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/edadataviz.ipynb

EDA with SQL

We utilized SQL queries for in-depth exploration of the SpaceX launch data. This enabled precise analysis, extracting targeted insights crucial for our first-stage landing prediction model.

- Key SQL Analyses & Insights:
 - Aggregations: Calculated total and average payload masses, and summarized mission outcomes.
 - Filtering: Identified specific launch sites, first successful landing dates, and detailed failed landing attempts.
 - Advanced Queries: Determined booster versions carrying maximum payloads and analyzed landing outcomes over time.
- Source Code:

https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

We created an interactive Folium map to visualize SpaceX launch sites and their surrounding infrastructure.

- Map Features & Insights:

- Launch Sites: Marked with interactive circles and markers, providing location context.
- Launch Outcomes: Color-coded markers (green/red) at each site showed success/failure rates.
- Proximity Analysis: Lines linked sites to railways, highways, and coastlines, displaying distances.

- Source Code:

https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

We developed a dynamic Plotly Dash dashboard to provide interactive insights into SpaceX launch records, enabling stakeholders to explore data on launch success and payload characteristics.

- Plot:
 - Launch Success Pie Chart: Displays the total successful launches across all sites or for a specific launch site.
 - Payload vs. Launch Outcome Scatter Plot: Visualizes the correlation between payload mass and launch success that can be filtered using payload range slider.
- These interactive elements facilitate:
 - Quick Overviews: Instantly see overall and site-specific success rates.
 - Granular Analysis: Explore how payload mass influences launch success, identifying optimal ranges or booster performance.
- Source Code:

https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/spacex_dash_app.py

Predictive Analysis (Classification)

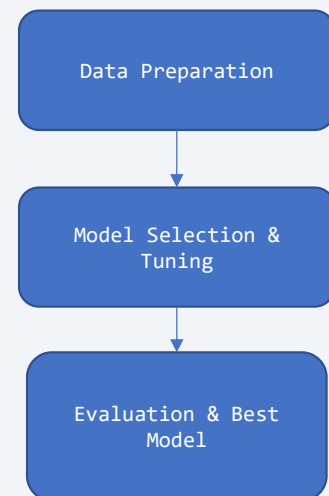
To predict SpaceX first-stage reusability, we developed and evaluated several classification models.

- **Model Development Process:**

- Data Preparation: Features scaled using StandardScaler and split into training/testing sets.
- Model Selection & Tuning: Applied Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN). Each model underwent GridSearchCV for hyperparameter tuning to optimize performance.
- Evaluation & Best Model: Models were evaluated based on classification accuracy. The Decision Tree Classifier emerged as the best-performing model, though all models showed comparable accuracy.

- **Source Code:**

https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb



Results

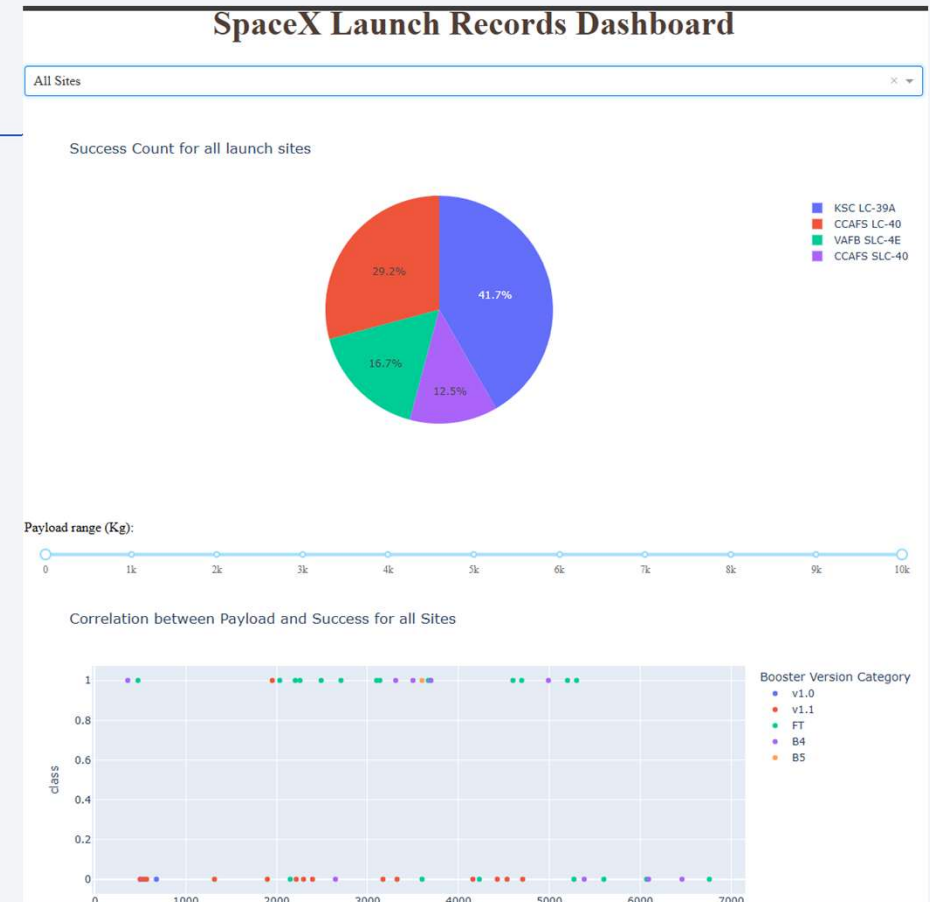
Our comprehensive analysis of SpaceX launch data yielded crucial insights across Exploratory Data Analysis, Interactive Analytics, and Predictive Modeling, directly supporting Space Y's mission to optimize operations and pricing.

- Exploratory Data Analysis Results:

- Launch Success Trends: Overall success rate for first-stage landings significantly increased with higher flight numbers and over time.
- Site Performance: Launch sites like CCAFS SLC 40 and KSC LC 39A demonstrated consistently higher success rates across various payload masses compared to VAFB SLC 4E.
- Orbit-Specific Outcomes: Orbits such as LEO, ISS, ES-L1, SSO, and VLEO showed high success, while GTO success improved over time.
- Payload Influence: Even with more massive payloads, the first stage often returned successfully, indicating that payload mass is a factor, but not a sole determinant for failure in later flights.
- Key Factors: Identified critical features highly correlated with landing success, including FlightNumber, PayloadMass, GridFins, Reused, Legs, Block, and ReusedCount.
- Logistical Insights: Mapped launch sites with their proximities to railways, highways, and coastlines to assess logistical advantages and environmental factors.

Results

- Predictive Analysis Results:
 - Model Performance: After hyperparameter tuning with GridSearchCV, Logistic Regression, SVM, Decision Tree, and K-Nearest Neighbors models all achieved strong predictive accuracy.
 - Best Performer: The Decision Tree Classifier exhibited a slightly higher accuracy of approximately 88.75%, demonstrating robust performance in predicting landing outcomes.

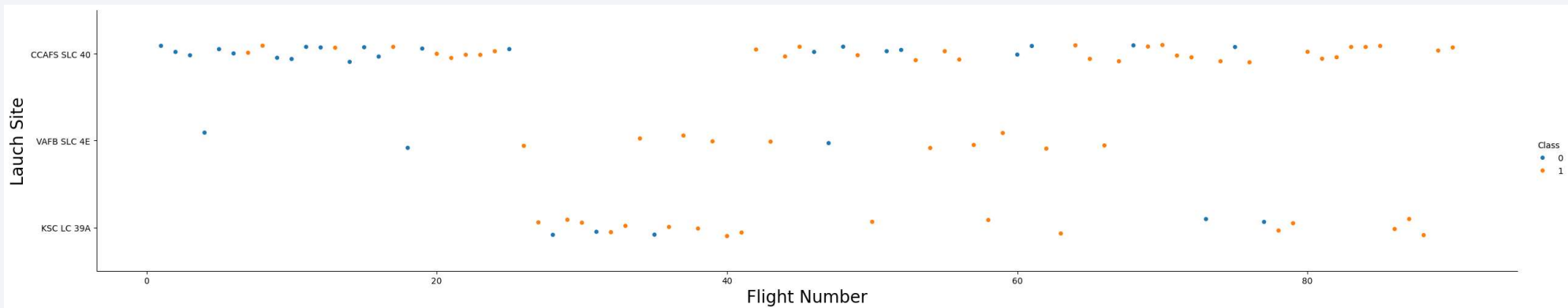


The background of the slide is a dynamic, abstract composition of numerous thin, overlapping lines and streaks in shades of blue, red, and teal. These lines are oriented diagonally, creating a sense of movement and depth. The overall effect is reminiscent of a high-speed data visualization or a complex network diagram.

Section 2

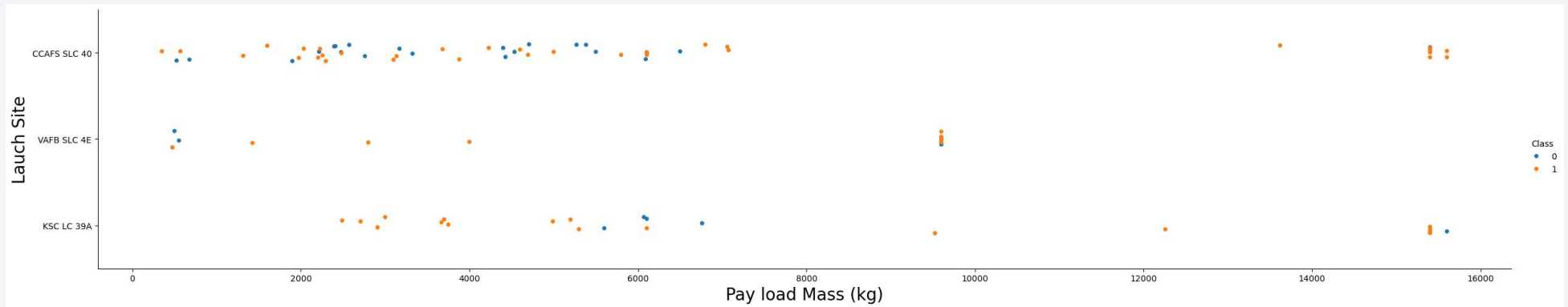
Insights drawn from EDA

Flight Number vs. Launch Site



Increased Success with Flight Number: For all launch sites, there's a general trend where successful landings become more frequent as the flight number increases. This suggests that SpaceX's landing capabilities improved over time.

Payload vs. Launch Site

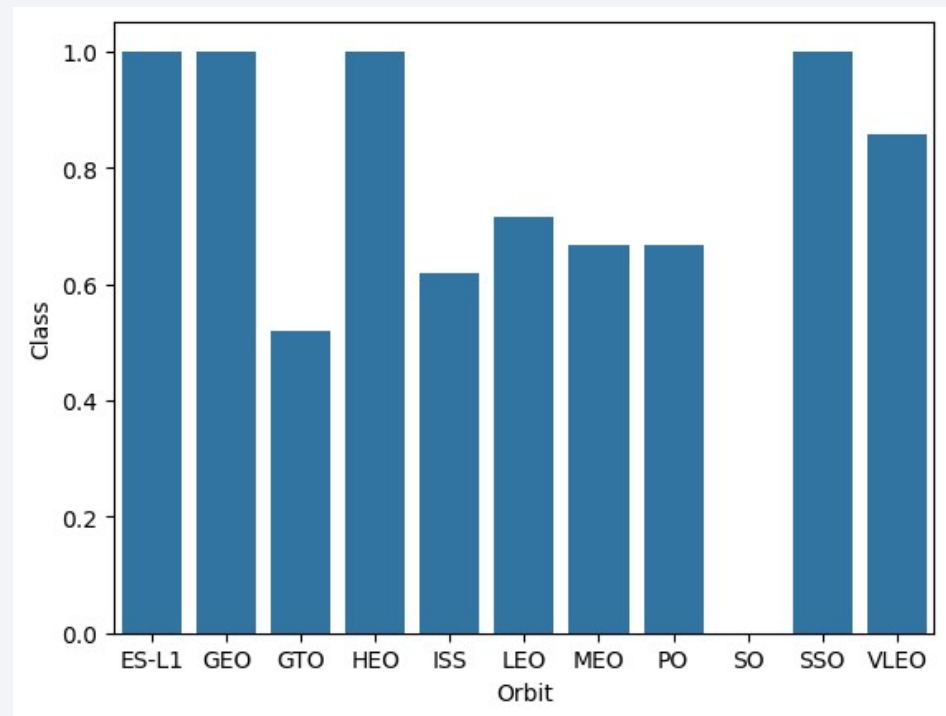


For the VAFB-SLC launch site there are no rockets launched for heavypayload mass (greater than 10000).

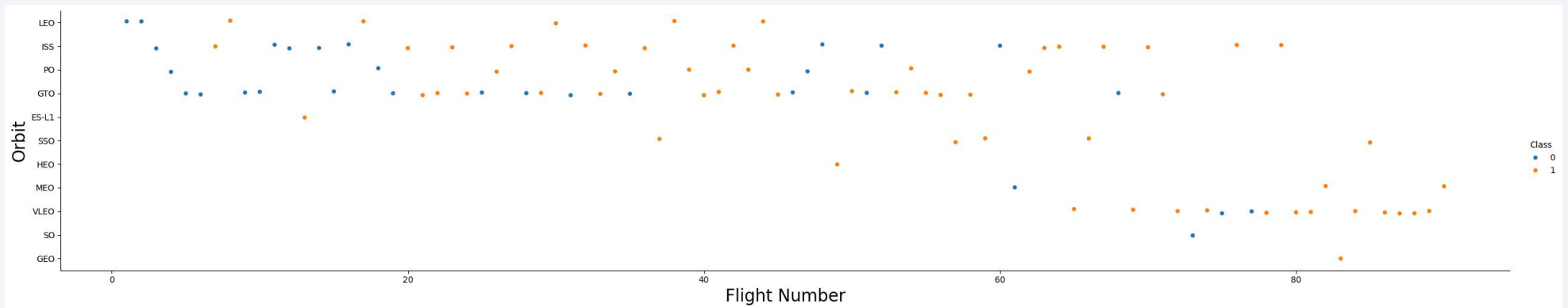
Success Rate vs. Orbit Type

The following orbits have the highest success rates:

- ES-L1
- GEO
- HEO
- SSO

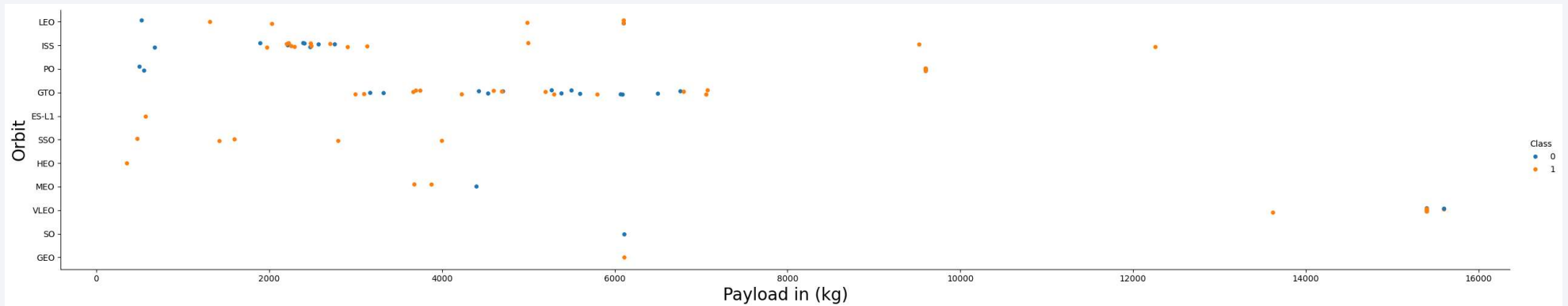


Flight Number vs. Orbit Type



In the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

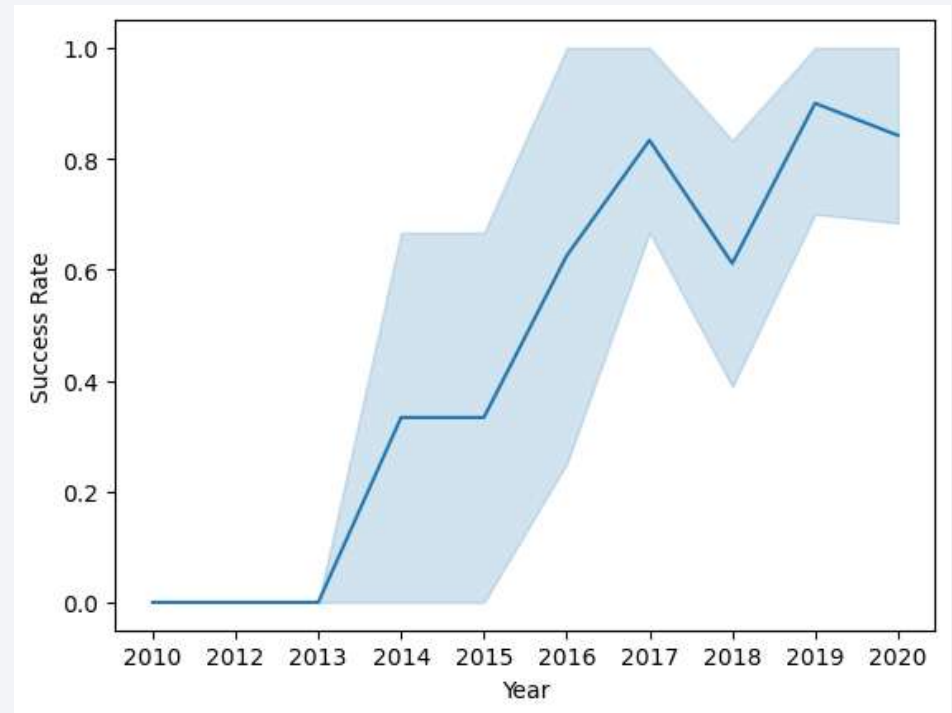
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS. However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend

The success rate since 2013 kept increasing till 2020



All Launch Site Names

Names of the unique launch sites extracted from Database created from downloaded CSV file of SpaceX mission records

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Displaying 5 records where launch sites begin with the string 'CCA'

Total Payload Mass

The total payload carried by boosters from NASA

```
SUM(PAYLOAD_MASS__KG_)
45596
```

Average Payload Mass by F9 v1.1

Displaying average payload mass carried by booster version F9 v1.1

```
AVG(PAYLOAD_MASS_KG_)
2534.666667
```

First Successful Ground Landing Date

Find the date of the first successful landing outcome on ground pad

```
MIN(Date)
0 2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

The total number of successful and failure mission outcomes

Mission_Outcome	COUNT(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

The names of the booster which have carried the maximum payload mass

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

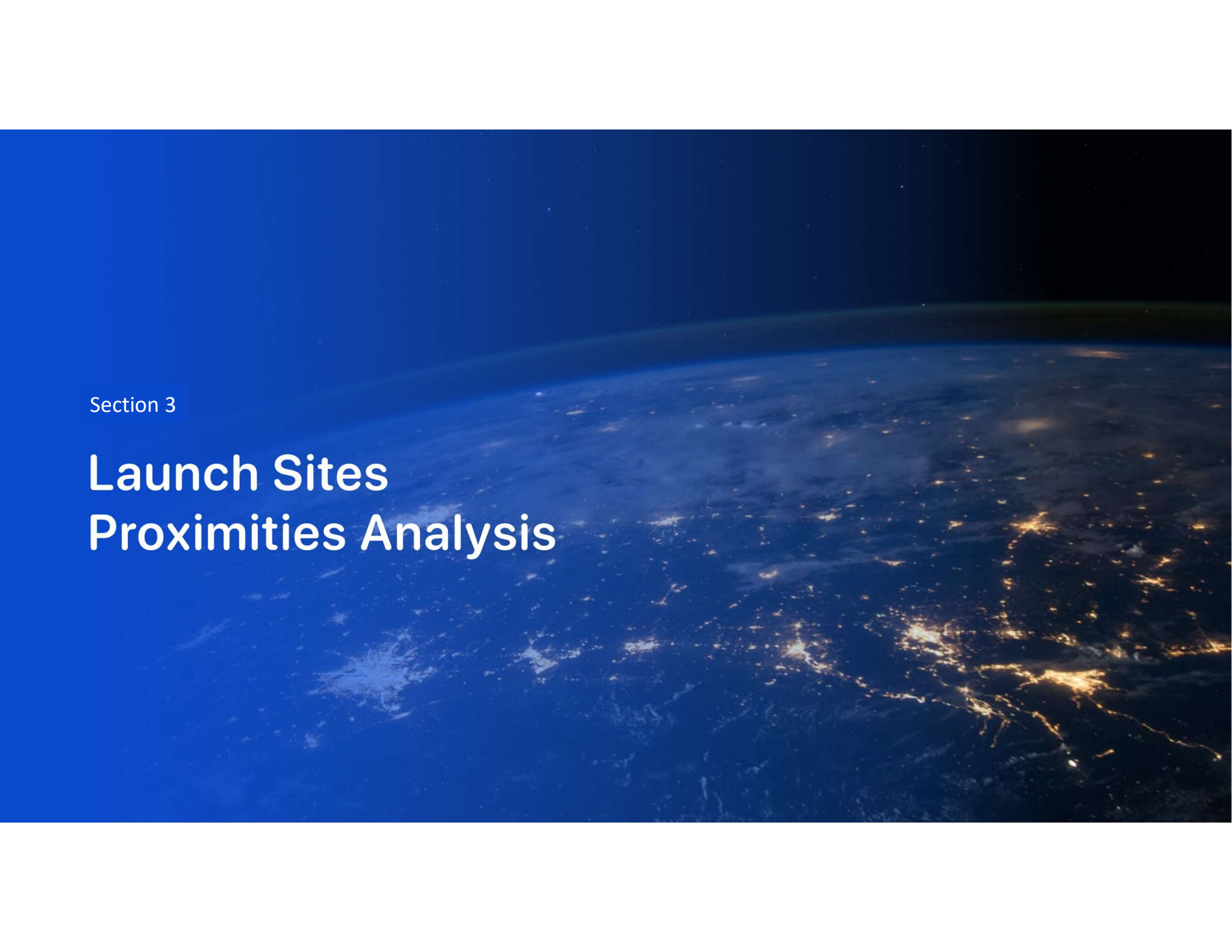
The records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Month_Name	Booster_Version	Launch_Site	Landing_Outcome
January	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
April	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Ranking of the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	Landing_Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is used as a background for the title slide.

Section 3

Launch Sites Proximities Analysis

Map of All launch sites

Most launch sites are concentrated in the United States, near the Florida coast and California.



Map with Color labeled launch outcomes

Cape Canaveral sites (KSC LC 39A and CCAFS SLC 40):

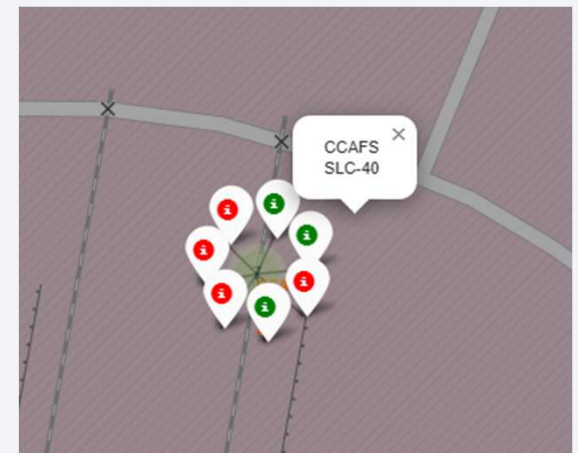
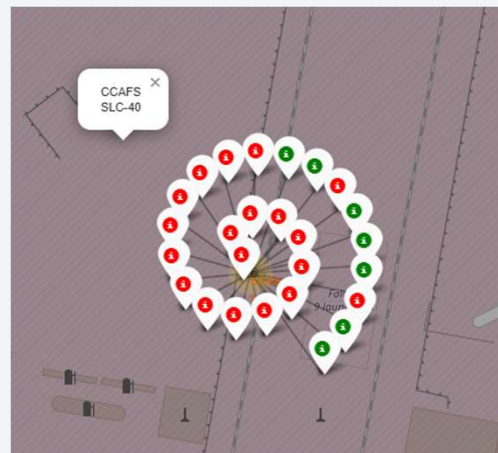
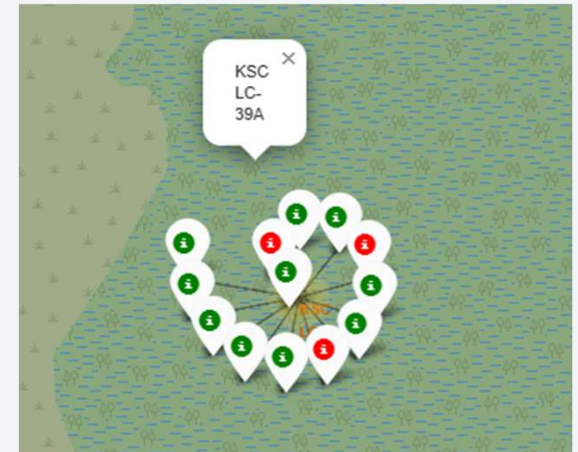
- High density of successful landings (green), confirming their reliability and frequency.
- VAFB SLC 4E (California):
- Fewer total launches.
- Some mixed outcomes, reflecting earlier-stage tests or missions with more risk.

CCAFS LC-40 and VAFB SLC-4E:

- More recent activity, with experimental launches.
- Shows some unsuccessful landings (red), which aligns with its role in early testing of Starship/Super Heavy.

Visual Insight:

- The color distribution clearly shows SpaceX's progress over time, with more recent missions increasingly resulting in successful landings.
- Geographical distribution of outcomes can help identify which sites have higher mission success rate



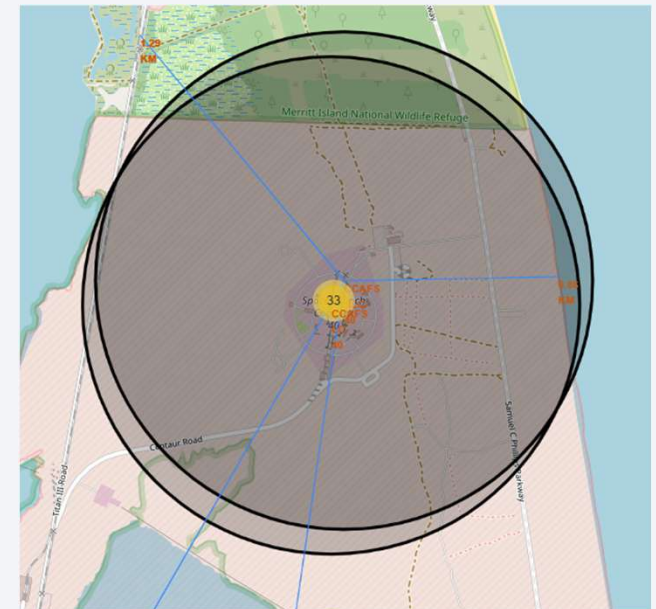
Mapping Nearby Infrastructure for Launch Operations

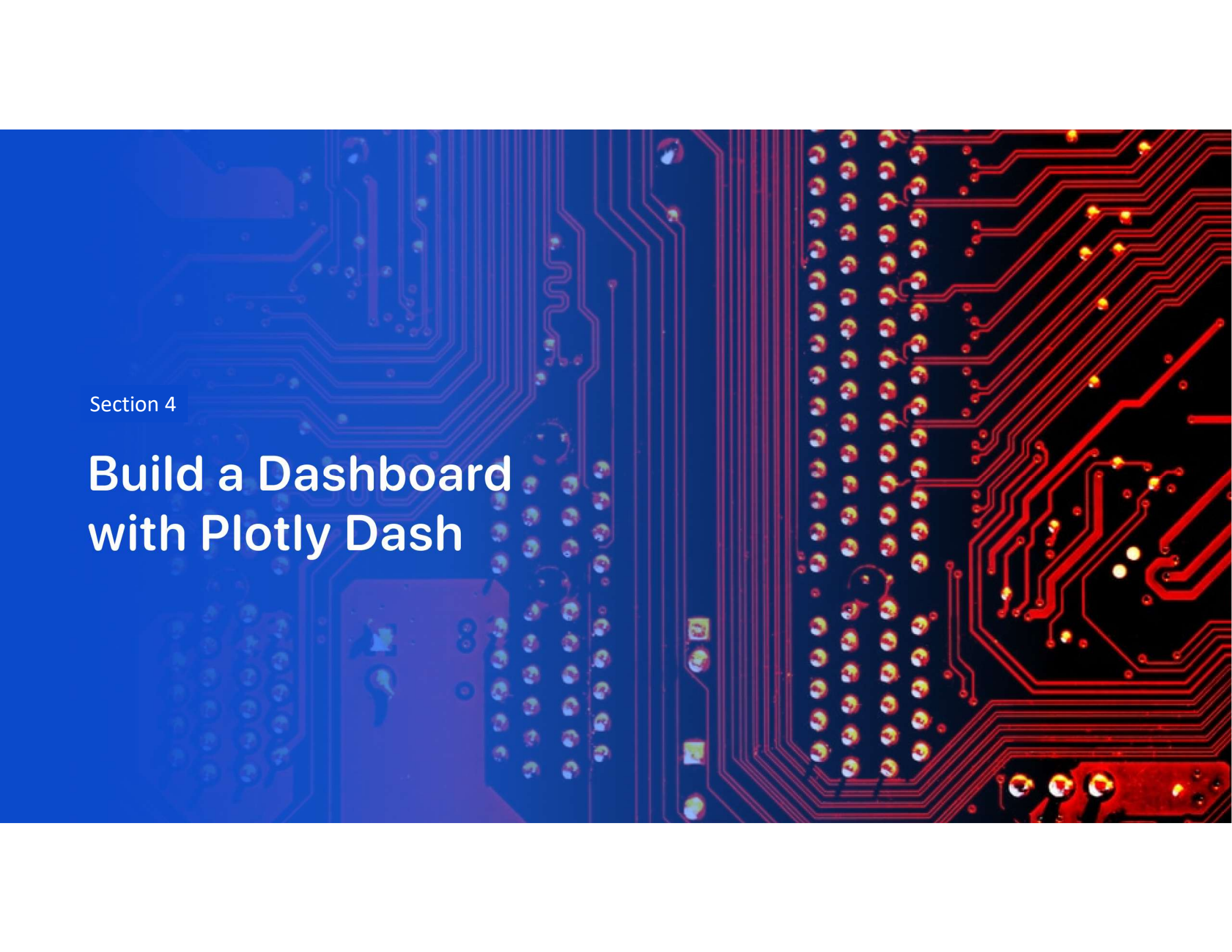
Close Proximity to Coastline:

- The launch site is located very near the coastline, confirming that launches over the ocean reduce risk to populated areas.

Accessibility to Transportation Infrastructure:

- Railways and highways are within a few kilometers of the launch site, enabling efficient transport of heavy rocket components and fuel.
- Example: Distance to the nearest highway was measured around 2.27 km, showing strong ground logistics support.



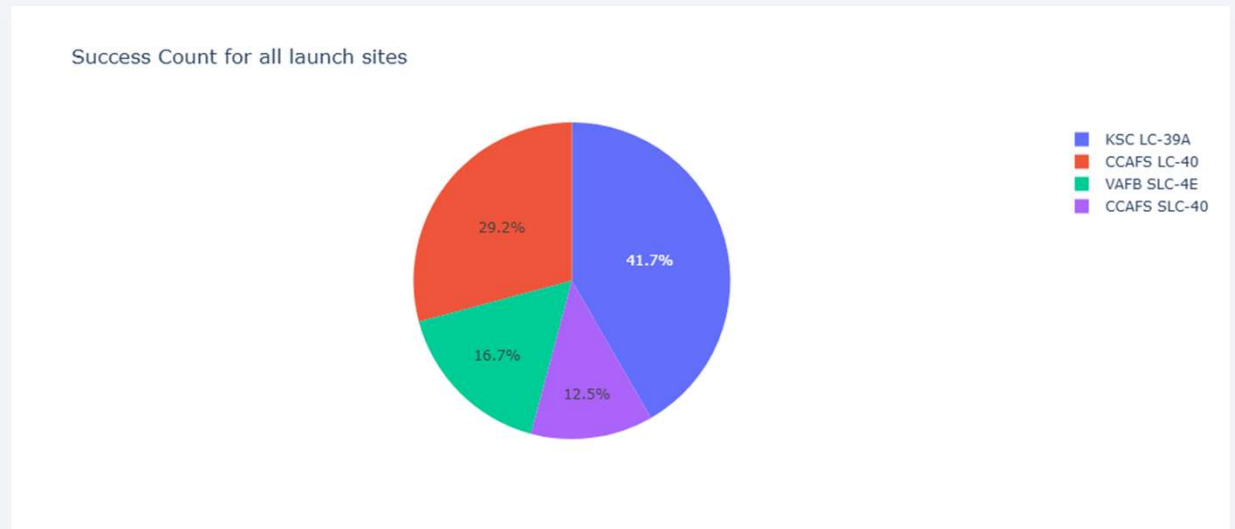


Section 4

Build a Dashboard with Plotly Dash

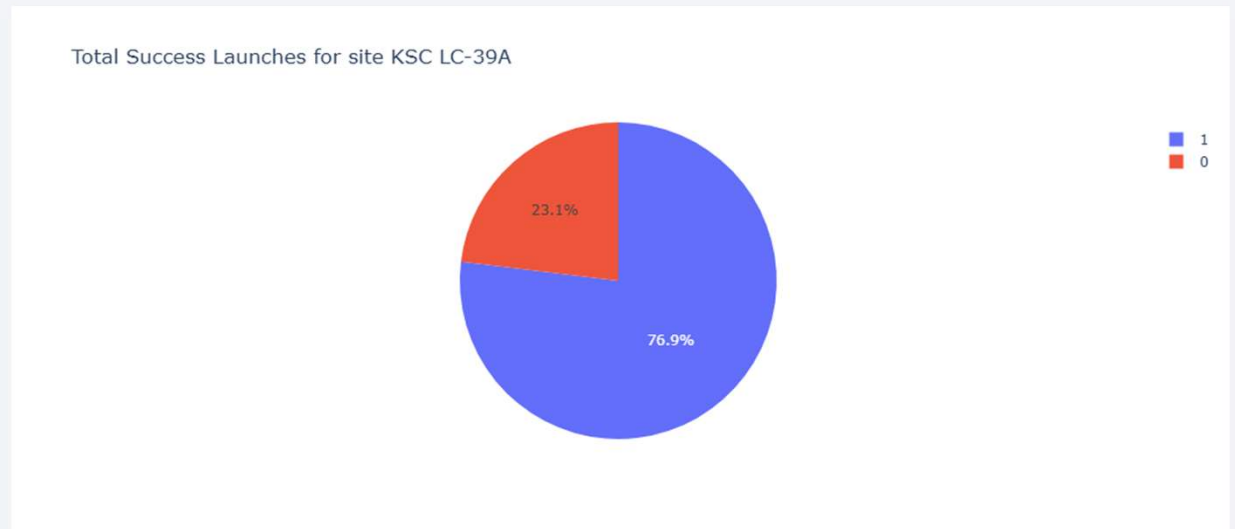
Success Count for all launch Sites

- KSC LC 39A has the highest number of successful launches.
- CCAFS LC 40 and VAFB SLC 4E follow in success count.
- Shows SpaceX's reliance on multiple launch sites and operational flexibility.



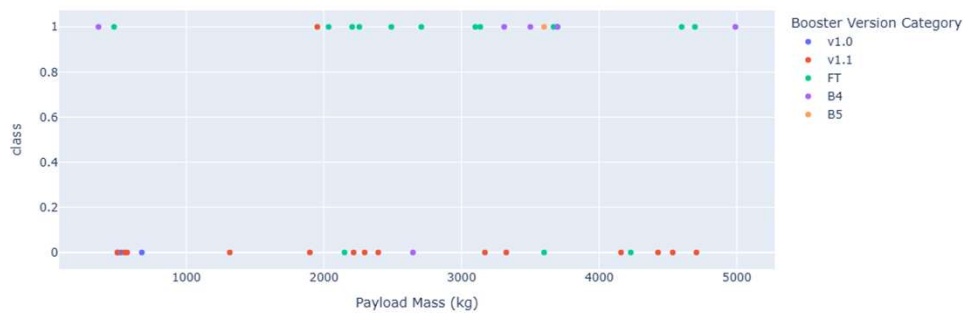
Success Count for KSC LC 39A

- KSC LC 39A has 77% success rate

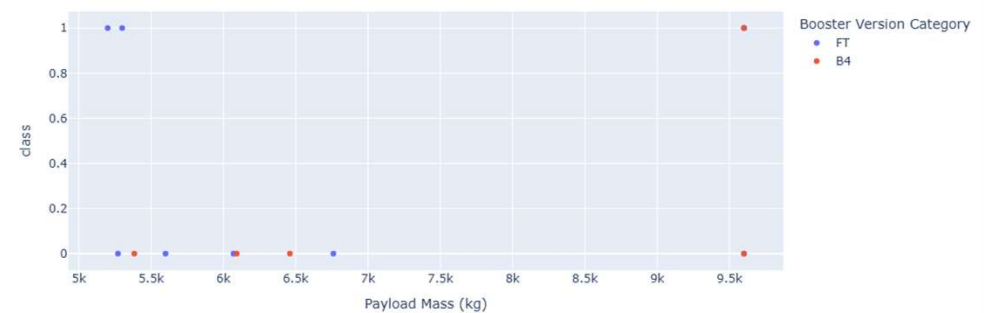


Payload vs Launch Outcome for all sites

Correlation between Payload and Success for all Sites



Correlation between Payload and Success for all Sites



- Success Trend: Launches with payloads between 2000–6000 kg show the highest success rate.
- Booster Performance: Most successful outcomes are linked to FT and B5 booster versions, which are newer and more reliable.
- Failure Zones: Launches with very low (<1000 kg) or very high (>8000 kg) payloads have a higher rate of failure or fewer launches.
- Launch Sites: Consistent success observed across all sites in the optimal payload range.
- Insight: Optimizing payload mass and using newer booster versions significantly increase the likelihood of successful launches.

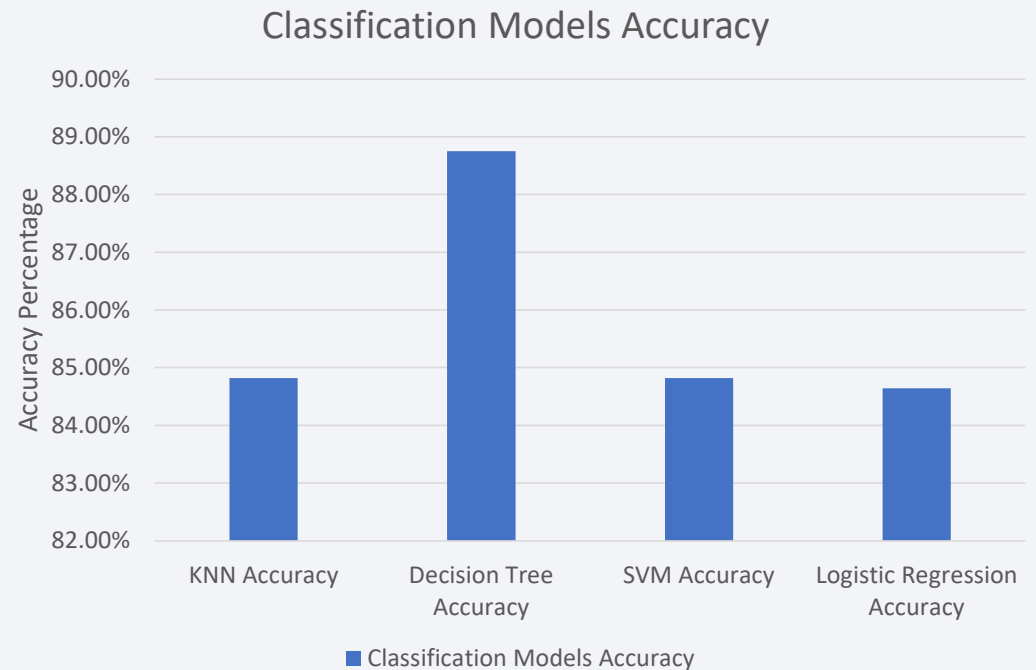
The background of the slide features a dynamic, abstract image. On the left, there is a solid blue area. To the right, a tunnel-like structure is depicted with curved, flowing lines in shades of blue and white, creating a sense of motion and depth. The overall aesthetic is modern and technological.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

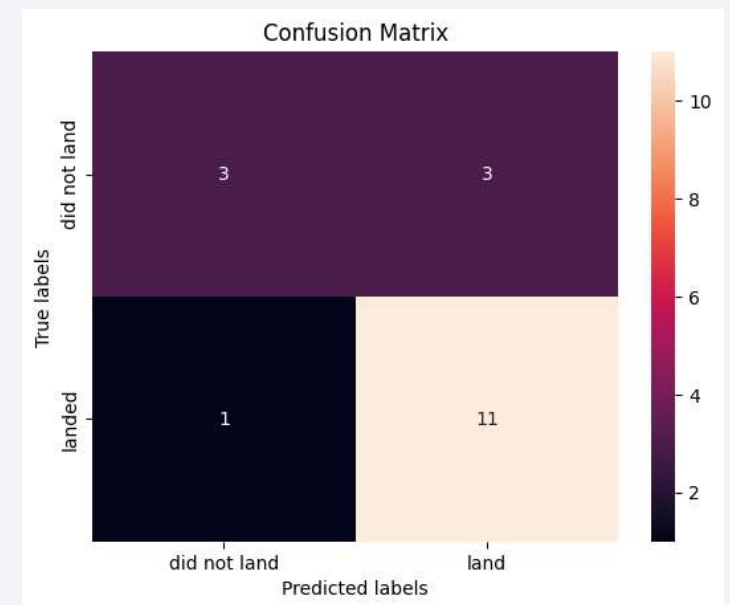
- After evaluating all the classification models, most demonstrated comparable accuracy. However, the **Decision Tree classifier** showed a slightly higher performance, making it the top-performing model among those tested.



Confusion Matrix

The confusion matrix for the Decision Tree model reveals that the classifier performs well at predicting both successful and failed first stage landings.

- True Positives (Success predicted as Success): High count, showing the model is good at identifying successful landings.
- True Negatives (Failure predicted as Failure): Also well captured, indicating good distinction.
- False Positives/Negatives: Low numbers, suggesting minimal misclassification.
- **Insight:** The model maintains a good balance between precision and recall, making it reliable for deployment in reuse prediction scenarios.



Conclusions

- **Comprehensive Analysis:** We successfully collected, cleaned, and explored SpaceX launch data using API calls, web scraping, and various visualization techniques.
- **Geospatial Insights:** Folium maps revealed spatial relationships between launch sites and nearby infrastructure (coastlines, railways, highways).
- **Interactive Dashboards:** Dash apps enabled user-driven analysis of payload effects, launch success trends, and site-specific performance.
- **Predictive Modeling:** Machine learning models were built and evaluated to predict first-stage landing outcomes, with Decision Tree classifier achieving the best performance.
- **Key Finding:** KSC LC-39A has the highest launch success rate; missions with mid-range payloads and specific booster versions show better success outcomes.

Appendix

- Github links to all files created used in this project
 - https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/jupyter-labs-spacex-data-collection-api.ipynb
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 - https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/spacex_dash_app.py
 - https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb
 - https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/dataset_part_1.csv
 - https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/dataset_part_2.csv
 - https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/dataset_part_3.csv
 - https://github.com/rmilad/TestRepo/blob/2792431bb3931e8322ba0d8e8ca99e3fec856ccc/Applied_Data_Science_Capstone/spacex_web_scraped.csv

Thank you!

